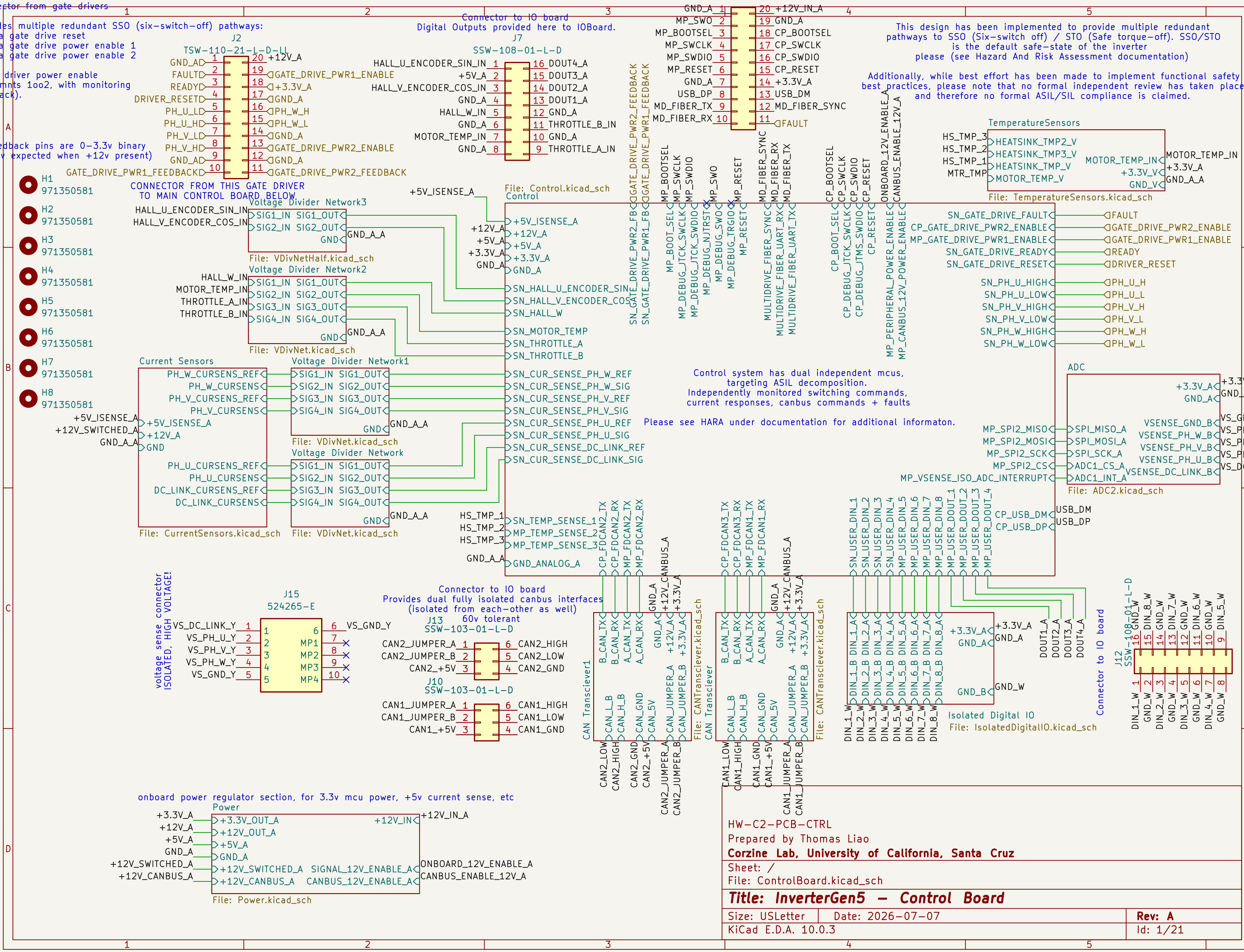
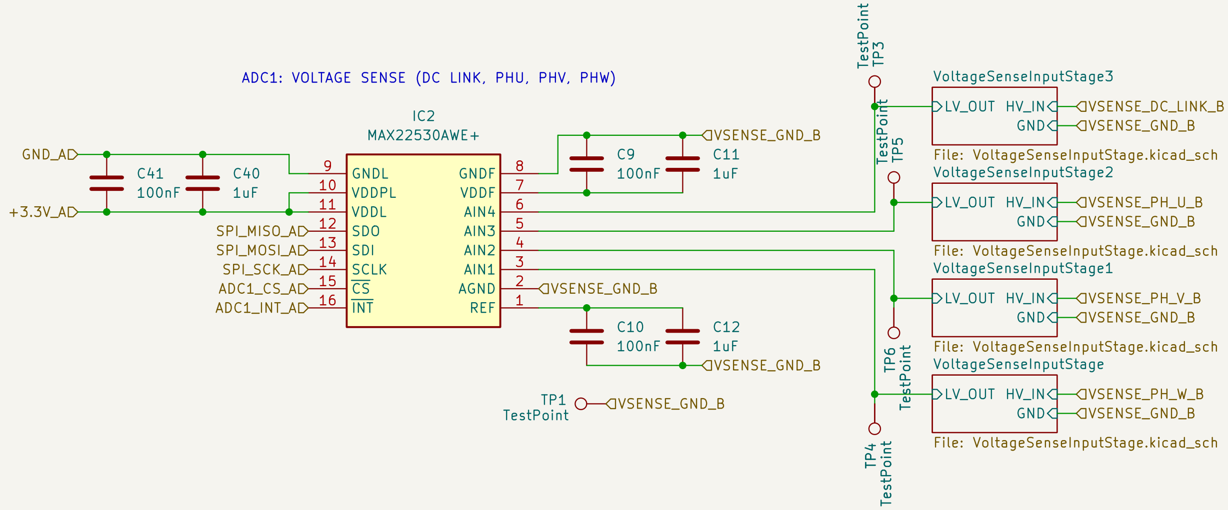


SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
J14
SSW-110-01-F-D
NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION,
DISABLE TIM1 SYNC PIN (TIM1 ETR)
Connector from gate drivers
Includes multiple redundant SSO (six-switch-off) pathways:
- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2
(Gate driver power enable
implemnts 1oo2, with monitoring
feedback).
feedback pins are 0-3.3v binary
(3.3v expected when +12v present)
J2
TSW-110-21-L-D-LL
CONNECTOR FROM THIS GATE DRIVER
TO MAIN CONTROL BOARD BELOW
Connector to IO board
Digital Outputs provided here to IOBoard.
J7
SSW-108-01-L-D
This design has been implemented to provide multiple redundant
pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO
is the default safe-state of the inverter
please (see Hazard And Risk Assessment documentation)
Additionally, while best effort has been made to implement functional safety
best practices, please note that no formal independent review has taken place,
and therefore no formal ASIL/SIL compliance is claimed.
TemperatureSensors
File: TemperatureSensors.kicad_sch
Control
File: Control.kicad_sch
Voltage Divider Network3
File: VDivNetHalf.kicad_sch
Voltage Divider Network2
File: VDivNet.kicad_sch
Current Sensors
File: CurrentSensors.kicad_sch
Voltage Divider Network1
File: VDivNet.kicad_sch
Voltage Divider Network
File: VDivNet.kicad_sch
H1 971350581
H2 971350581
H3 971350581
H4 971350581
H5 971350581
H6 971350581
H7 971350581
H8 971350581
Control system has dual independent mcus,
targeting ASIL decomposition.
Independently monitored switching commands,
current responses, canbus commands + faults
Please see HARA under documentation for additional informaton.
ADC
File: ADC2.kicad_sch
voltage sense connector
ISOLATED, HIGH VOLTAGE!
J15
524265-E
Connector to IO board
Provides dual fully isolated canbus interfaces
(isolated from each-other as well)
60v tolerant
J13
SSW-103-01-L-D
J10
SSW-103-01-L-D
CAN Transciever1
File: CANTransciever.kicad_sch
CAN Transciever
File: CANTransciever.kicad_sch
Isolated Digital IO
File: IsolatedDigitalIO.kicad_sch
Connector to IO board
J12
SSW-108-01-L-D
onboard power regulator section, for 3.3v mcu power, +5v current sense, etc
Power
File: Power.kicad_sch
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: ControlBoard.kicad_sch
Title: InverterGen5 - Control Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/21

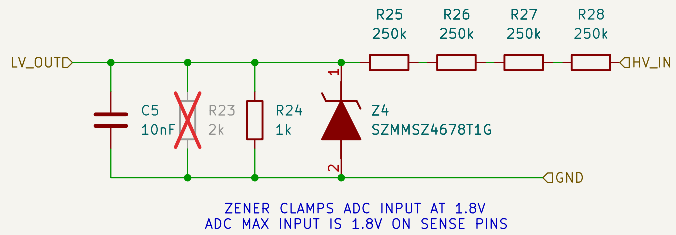


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/
File: ADC2.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 4/21

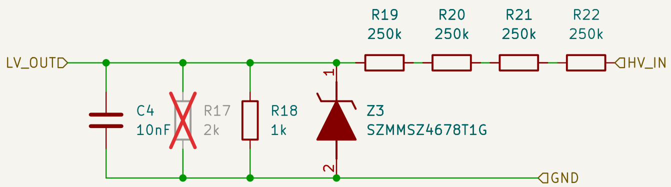


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage/
File: VoltageSenseInputStage.kicad_sch

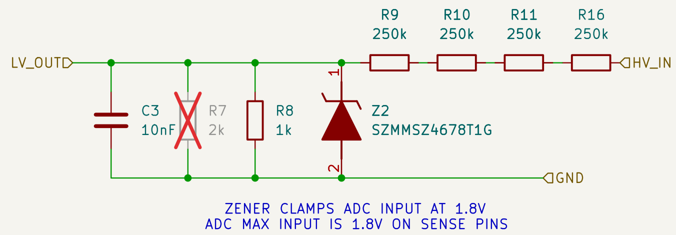
Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 2/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage1/
File: VoltageSenseInputStage.kicad_sch
Title: InverterGen5 — Control Board
Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 3/21

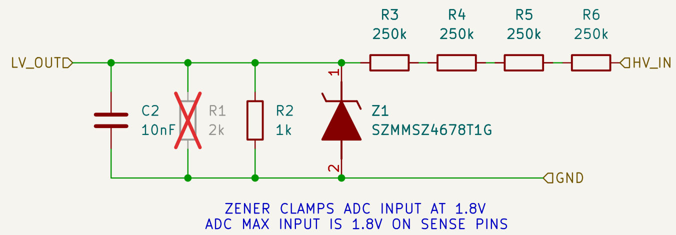


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/
File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 5/21

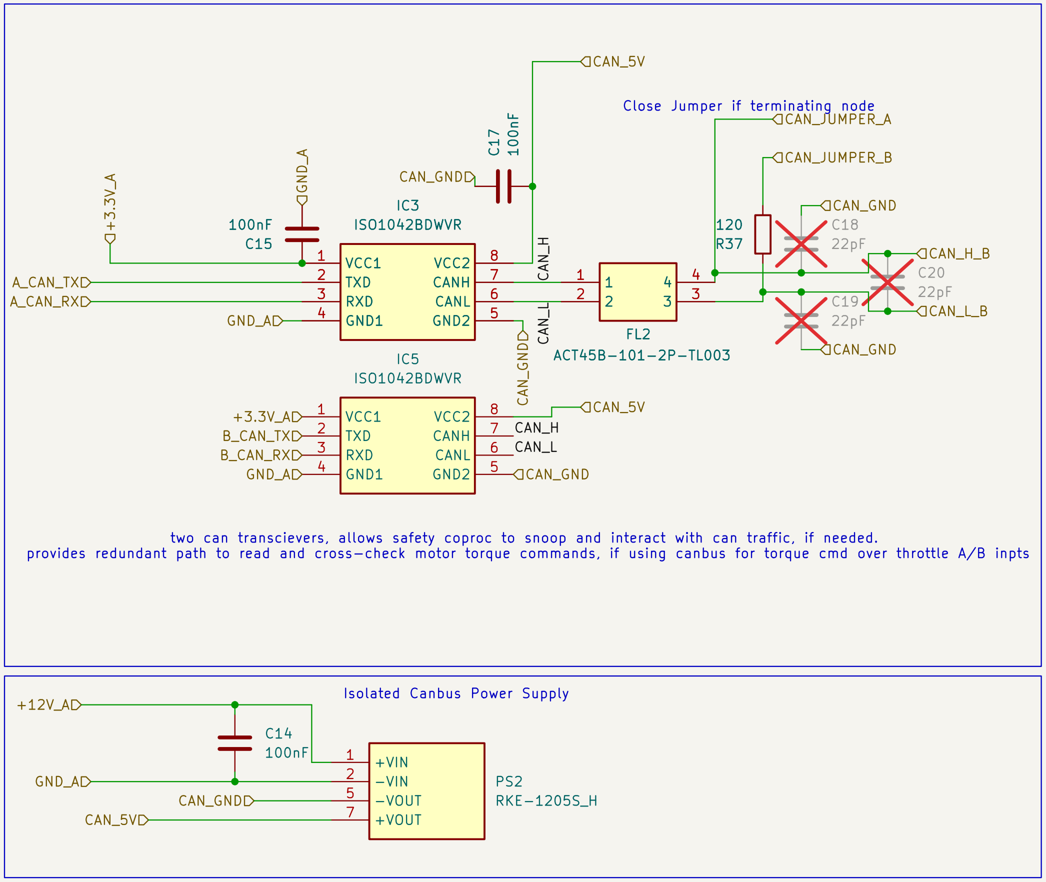


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/
File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 – Control Board

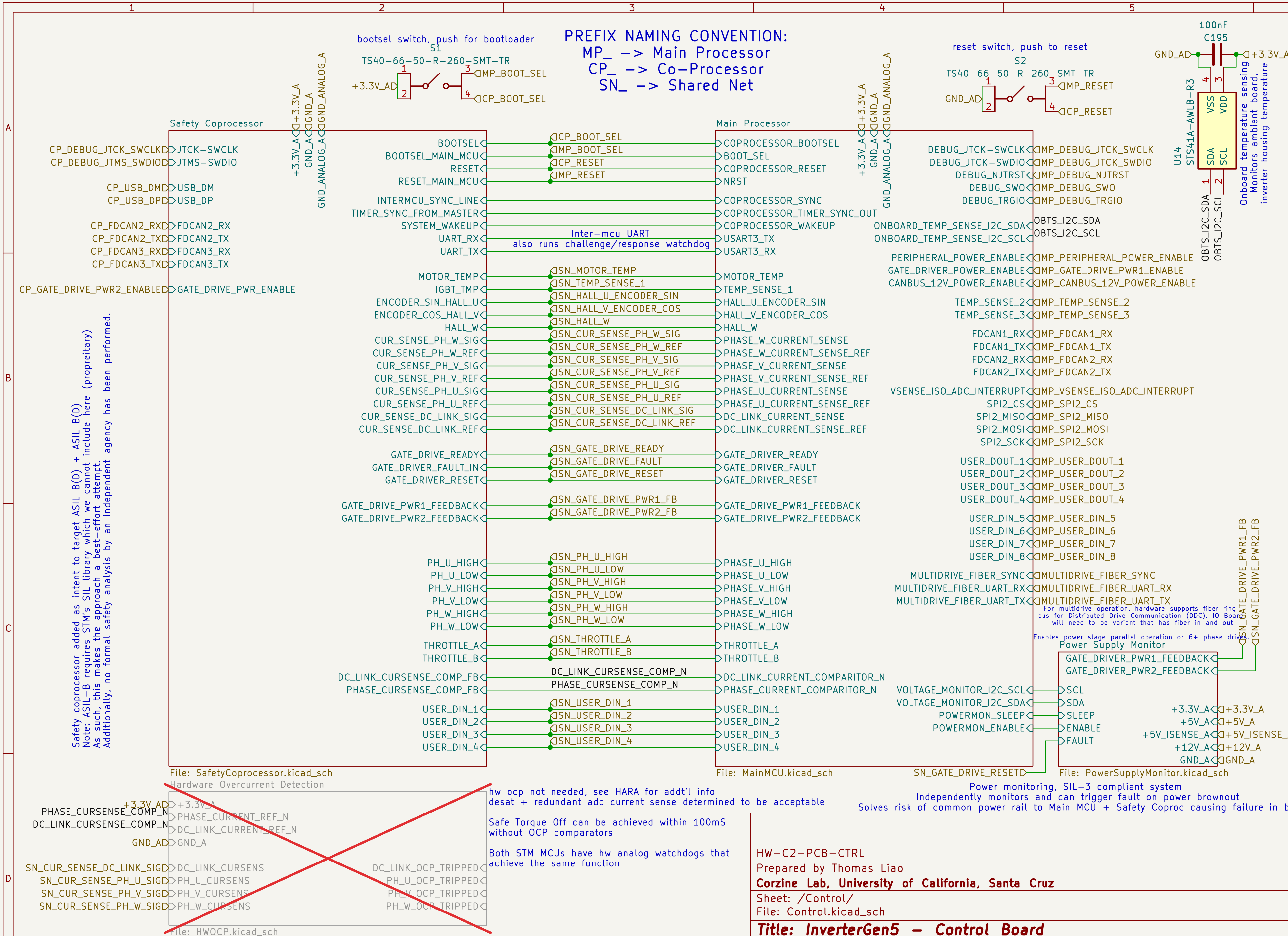
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 6/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever1/
File: CANTransciever.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 7/21

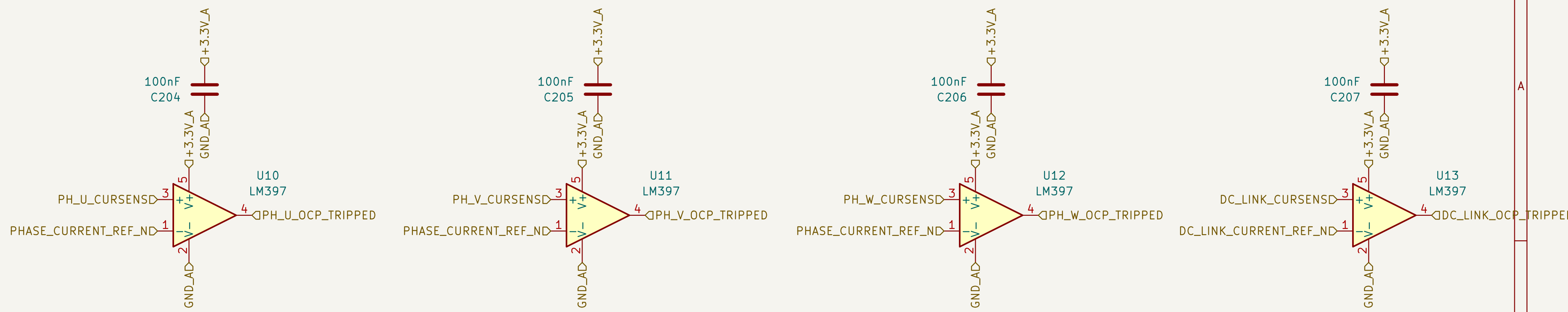


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/
File: Control.kicad_sch

Title: InverterGen5 – Control Board

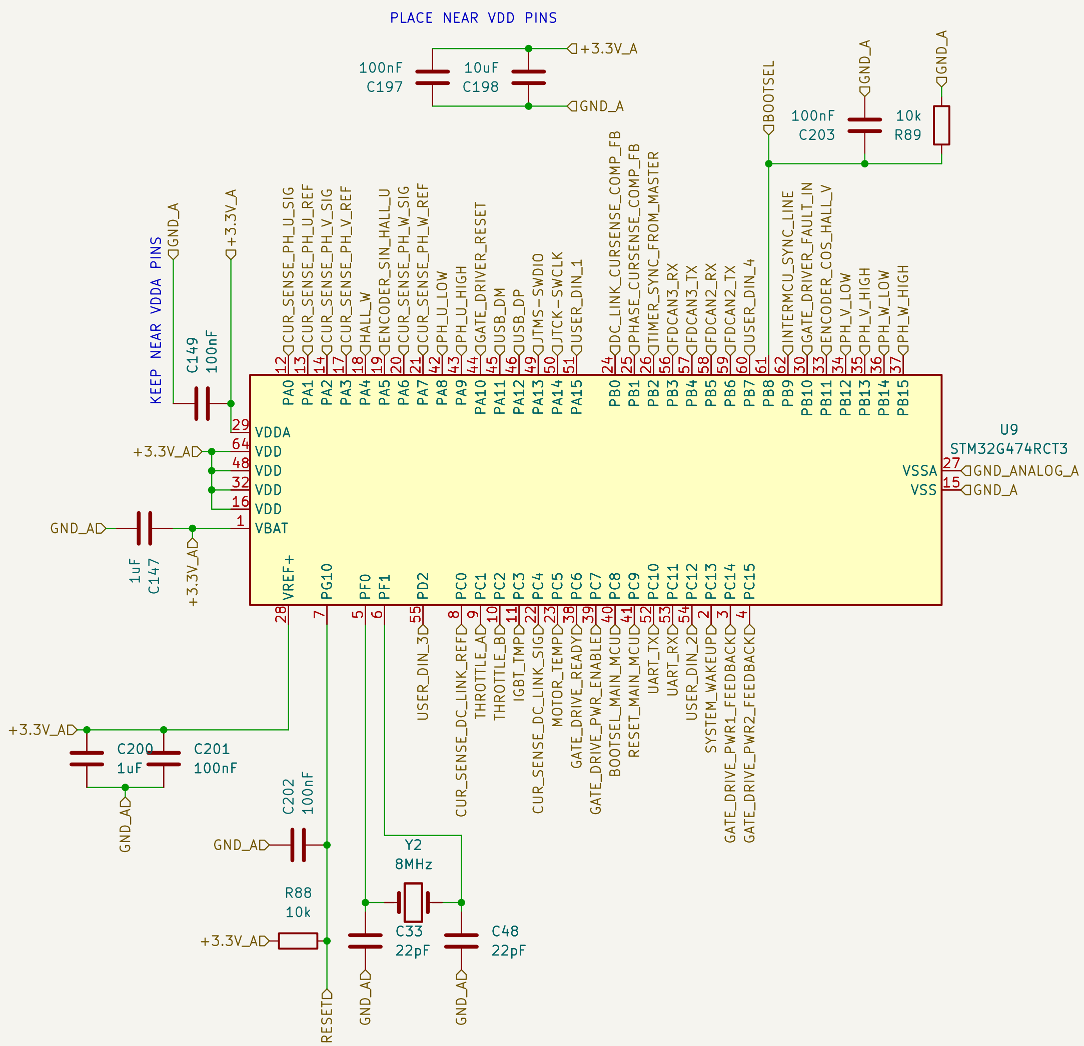
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 8/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Hardware Overcurrent Detection/
File: HWOCP.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 10/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

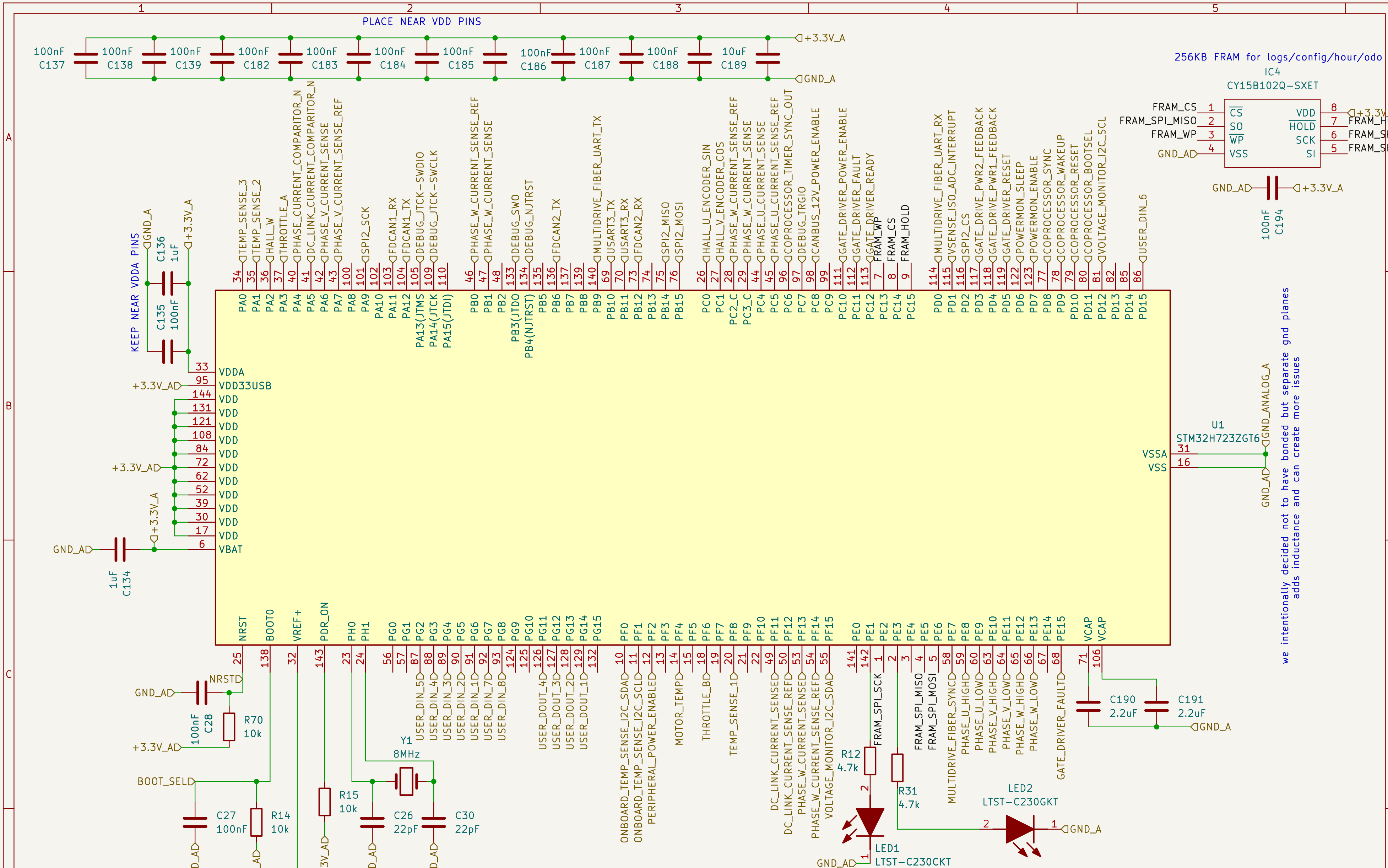
Sheet: /Control/Safety Coprocessor/

File: SafetyCoprocessor.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

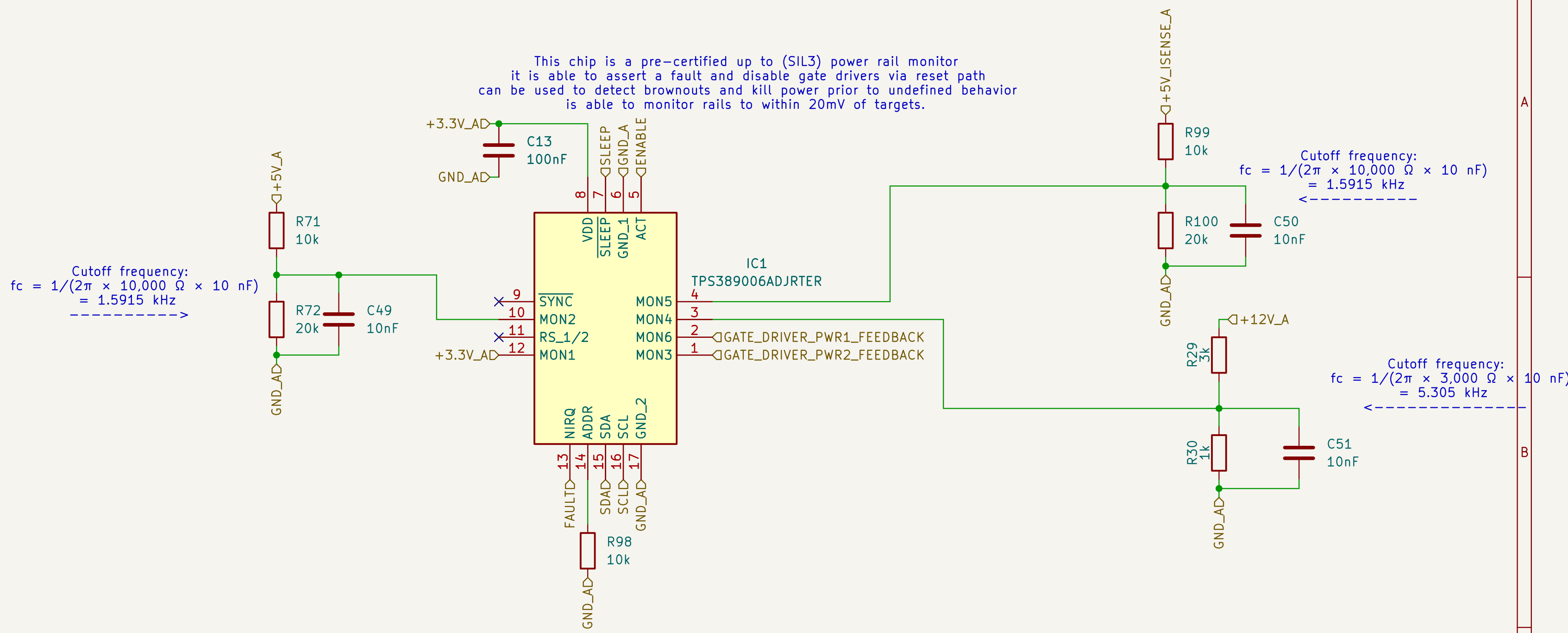
KiCad E.D.A. 10.0.3 | Id: 16/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Main Processor/
File: MainMCU.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 19/21



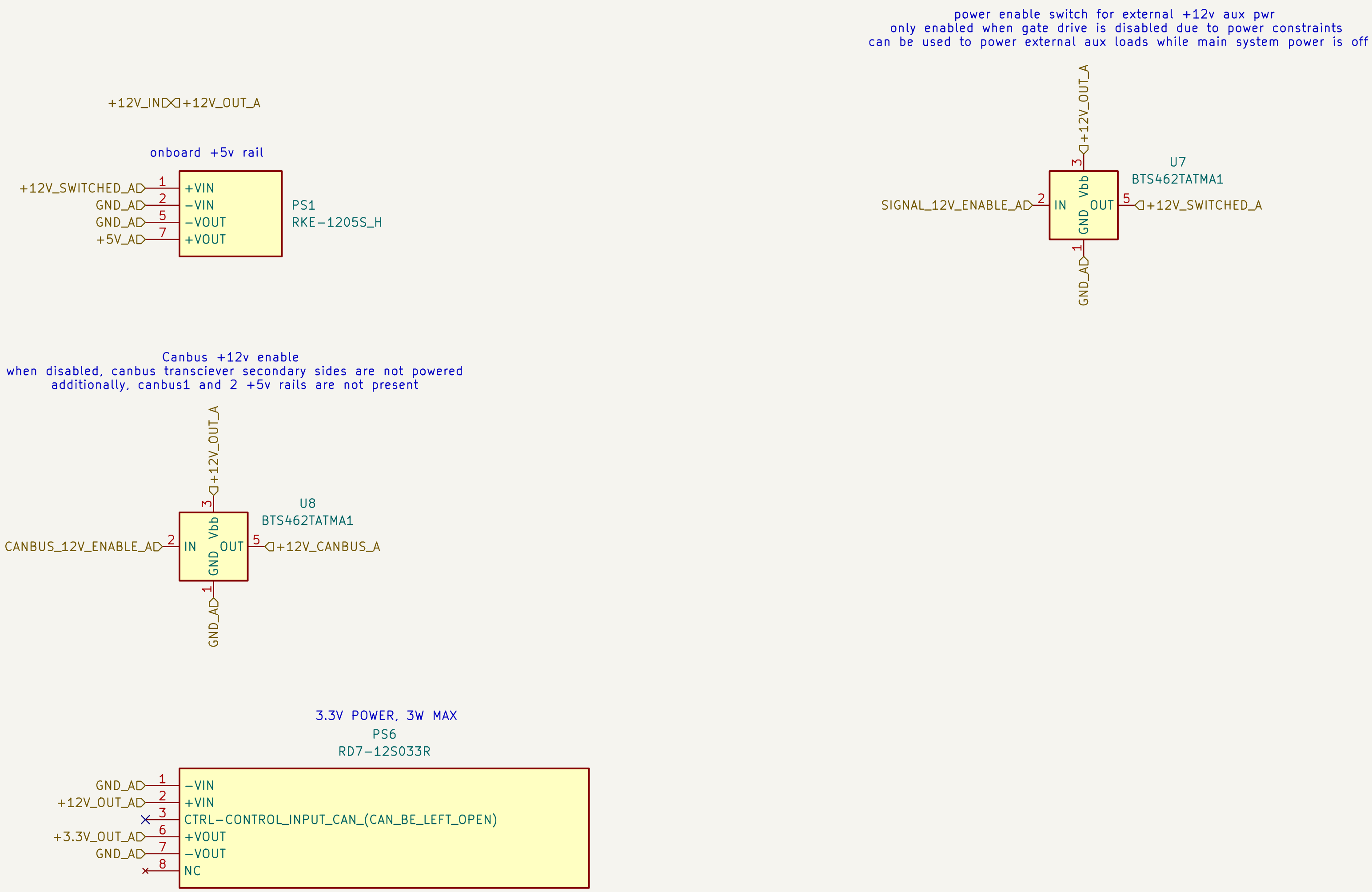
fault here is open collector, on error it pulls down
can be tied in parallel with pwm brkin faults from desat
will auto disable pwm on failure

HW–C2–PCB–CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Power Supply Monitor/
File: PowerSupplyMonitor.kicad_sch

Title: InverterGen5 – Control Board

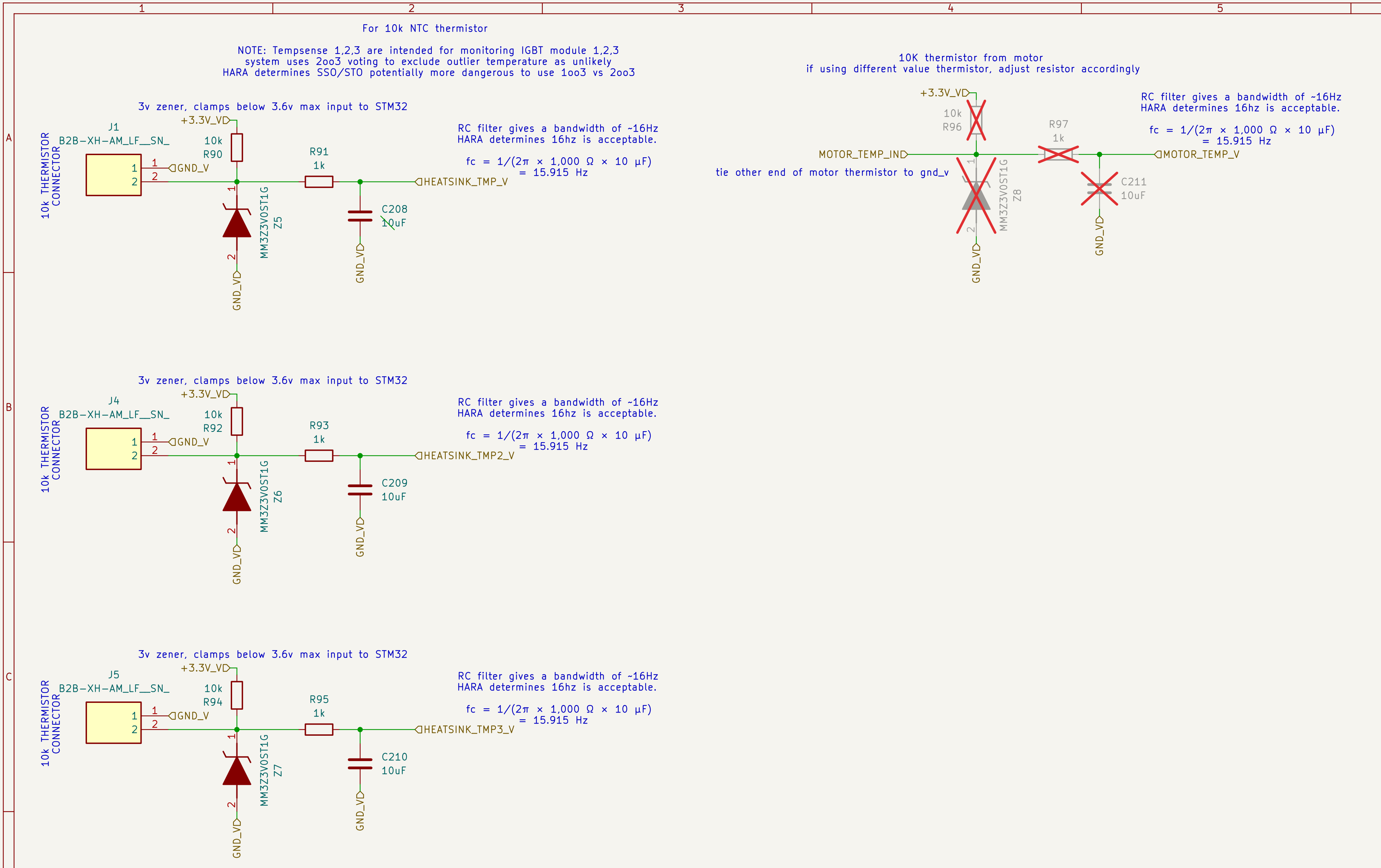
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 20/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Power/
File: Power.kicad_sch

Title: InverterGen5 - Control Board

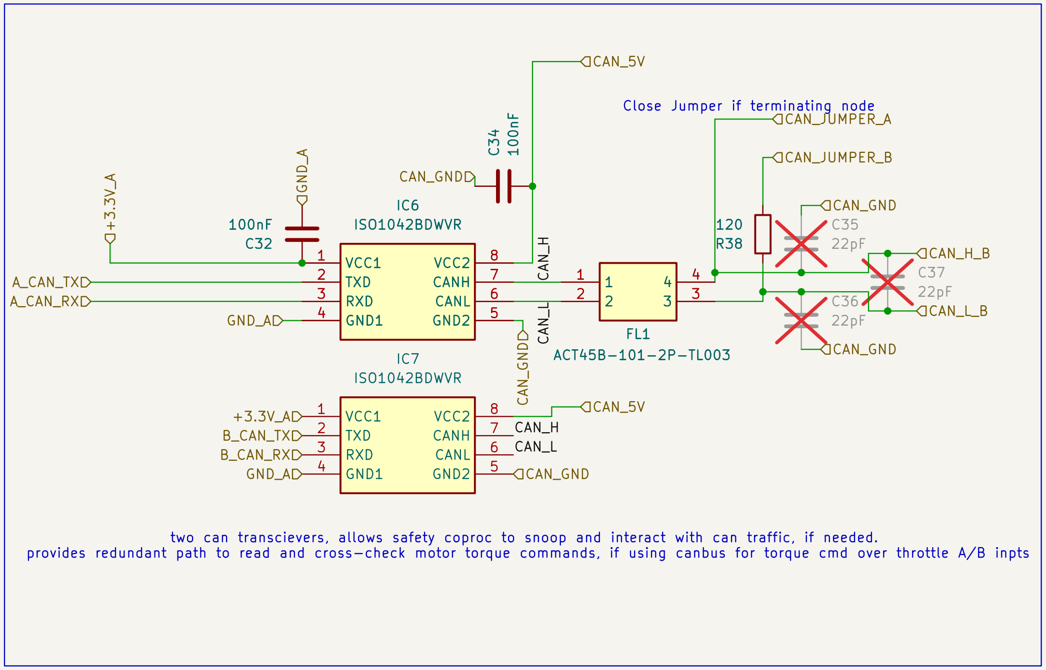
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 9/21

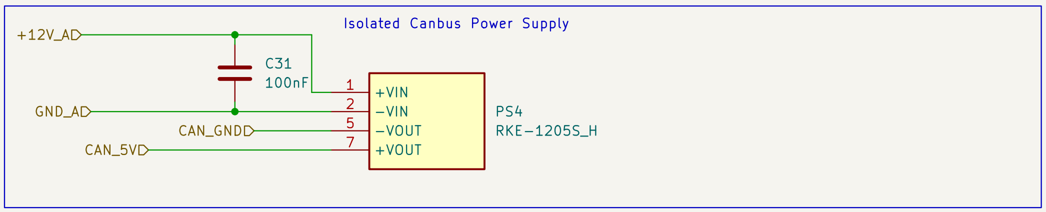


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /TemperatureSensors/
File: TemperatureSensors.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 9/21

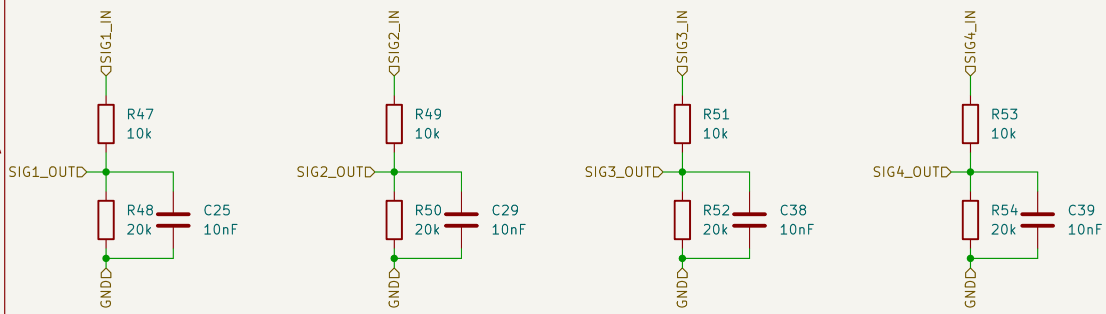




HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever/
File: CANTransciever.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 11/21

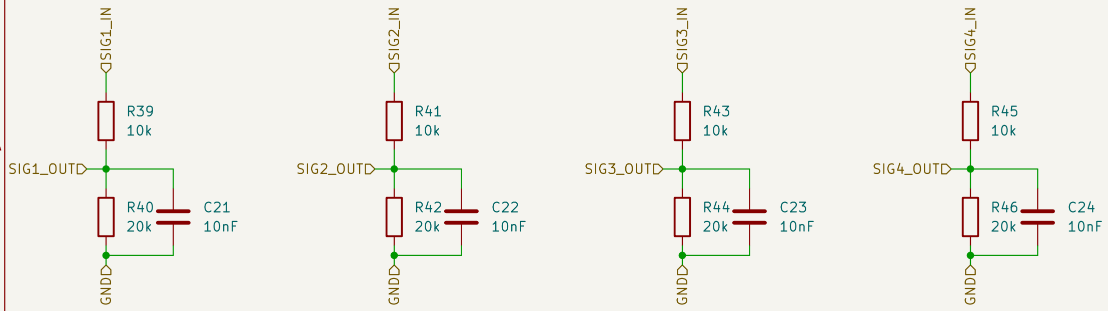


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/
File: VDivNet.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 12/21

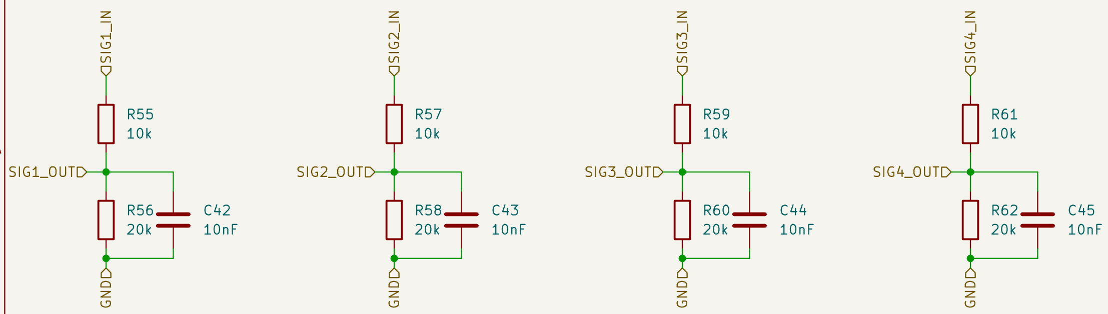


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network/
File: VDivNet.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 13/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

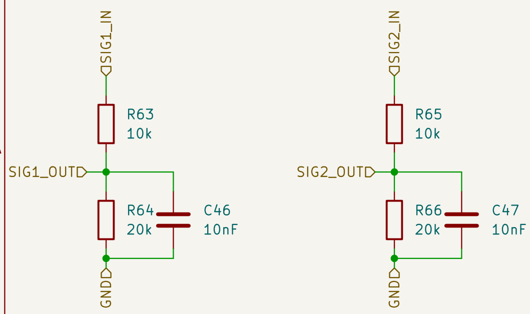
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network2/

File: VDivNet.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 14/21

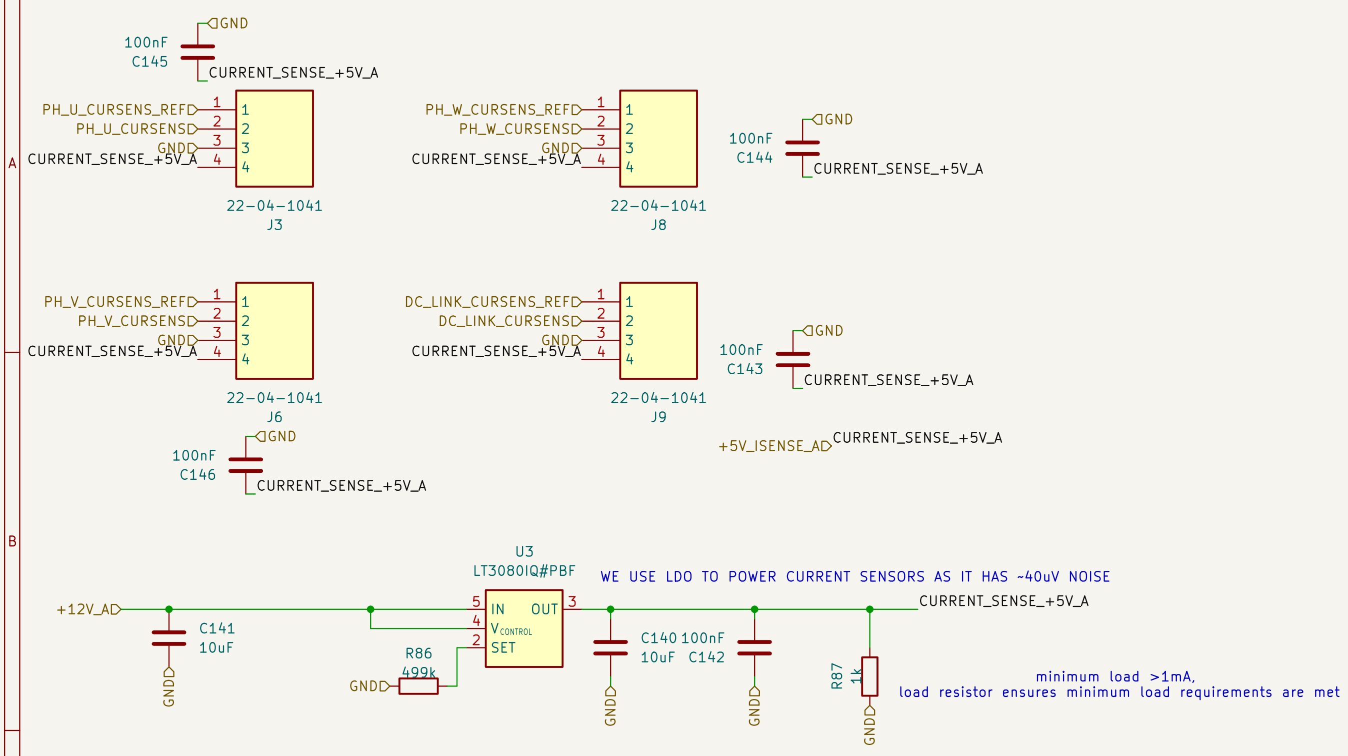


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network3/
File: VDivNetHalf.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 15/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

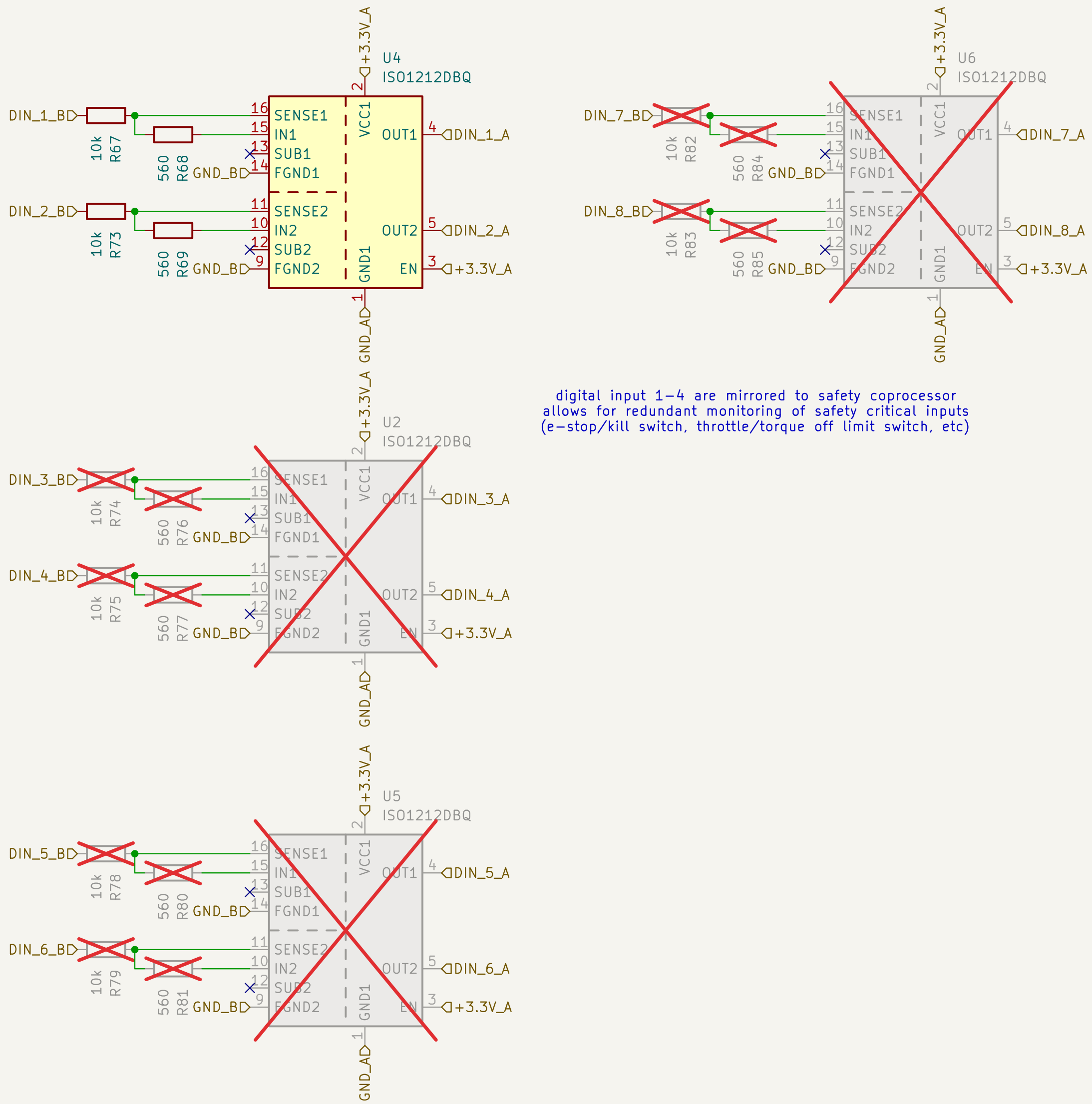
Corzine Lab, University of California, Santa Cruz

Sheet: /Current Sensors/

File: CurrentSensors.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 17/21



digital input 1-4 are mirrored to safety coprocessor
allows for redundant monitoring of safety critical inputs
(e-stop/kill switch, throttle/torque off limit switch, etc)

HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Isolated Digital IO/
File: IsolatedDigitalIO.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 18/21